

Alfentanil (United States and Canada: Not available): Drug information - UpToDate

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For additional information see "Alfentanil (United States and Canada: Not available): Patient drug information" and "Alfentanil (United States and Canada: Not available): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Special Alerts

Opioid Labeling Safety Updates August 2025

The FDA is requiring safety labeling changes for all opioid pain medications to better highlight the risks of long-term use, including addiction, misuse, and overdose. This decision follows a public advisory meeting that reviewed data from 2 large postmarketing studies, which revealed serious risks and a lack of evidence supporting long-term opioid effectiveness. The updated labels will include clearer risk information, stronger dosing warnings, revised guidance on appropriate use, details on overdose reversal agents, and drug interactions. They will also stress safe discontinuation to prevent harm, outline new risks like overdose-related brain injury and digestive issues, and strengthen warnings about interactions with CNS depressants, including gabapentinoids. Additionally, the FDA is requiring a new clinical trial to further assess long-term opioid use.

Further information may be found at: <https://www.fda.gov/drugs/drug-safety-and-availability/fda-requiring-opioid-pain-medicine-manufacturers-update-prescribing-information-regarding-long-term>

ALERT: US Boxed Warning

Addiction, abuse, and misuse:

Because the use of alfentanil exposes patients and other users to the risks of opioid addiction, abuse, and misuse, which can lead to overdose and death, assess each patient's risk prior to prescribing and reassess all patients regularly for the development of these behaviors and conditions.

Life-threatening respiratory depression:

Serious, life-threatening, or fatal respiratory depression may occur with use of alfentanil, especially during initiation or following a dosage increase. To reduce the risk of respiratory depression, proper dosing and titration of alfentanil are essential.

Risks from concomitant use with benzodiazepines or other CNS depressants:

Concomitant use of opioids with benzodiazepines or other CNS depressants, including alcohol, may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing

of alfentanil and benzodiazepines or other CNS depressants for use in patients for whom alternative treatment options are inadequate.

Cytochrome P450 3A4 interaction:

The concomitant use of alfentanil with all cytochrome P450 3A4 inhibitors may result in an increase in alfentanil plasma concentrations, which could increase or prolong adverse reactions and may cause potentially fatal respiratory depression. In addition, discontinuation of a concomitantly used cytochrome P450 3A4 inducer may result in an increase in alfentanil plasma concentration. Monitor patients receiving alfentanil and any CYP3A4 inhibitor or inducer.

Pharmacologic Category

- Analgesic, Opioid;
- Anilidopiperidine Opioid

Dosing: Adult

Dosage guidance:

Safety: Ensure an opioid overdose reversal agent (eg, naloxone, nalmefene), resuscitative and intubation equipment, and oxygen are readily available.

Dosing: Dosing should be individualized based on patient-specific factors (eg, comorbidities, severity of pain, degree of opioid experience/tolerance) and titrated to patient-specific treatment goals (eg, improvement in function and quality of life, decrease in pain using a validated pain rating scale). Use the lowest effective dose for the shortest period of time.

Dosage form information: Alfentanil has been discontinued in the United States for >1 year.

Analgesia or anesthesia

Analgesia or anesthesia: IV:

Weight-based dosing should utilize actual body weight for dosing calculations, unless >20% above ideal body weight, then use lean body weight for dosing calculations. Administer induction doses slowly over 3 minutes; as induction dosing may produce loss of vascular tone and hypotension, consider fluid replacement prior to induction. Refer to institutional protocols.

Alfentanil Dosage Range for Use During Anesthesia^a

Indication

Estimated duration of procedure

Induction period (initial dose)

Maintenance period (incremental injection)

Maintenance period (continuous infusion)

Total dose

Comments

^a Gropper 2019; manufacturer's labeling.

Analgesia (adjunctive agent)

≤30 minutes

8 to 20 mcg/kg

3 to 5 mcg/kg every 5 to 20 minutes

0.25 to 1 mcg/kg/minute

8 to 40 mcg/kg

Spontaneously breathing or assisted ventilation when required.

Analgesia or anesthesia (adjunctive agent)

30 to 60 minutes

20 to 50 mcg/kg (~10 mcg/kg for laryngeal mask airway when co-administered with propofol)

5 to 15 mcg/kg every 5 to 20 minutes

Up to 75 mcg/kg

May be administered in divided doses.

Assisted or controlled ventilation required.

Discontinue at least 30 minutes prior to end of surgery.

Anesthesia, induction (adjunctive agent)

>45 minutes

25 to 100 mcg/kg

Up to 245 mcg/kg

May be administered in divided doses.

Assisted or controlled ventilation required.

Concentration of volatile inhalation anesthetics reduced by 30% to 50% for initial hour.

Following an anesthetic induction dose of alfentanil, alfentanil infusion rate requirements are reduced by 30% to 50% for the first hour of maintenance.

Anesthesia, maintenance (adjunctive agent)

>45 minutes

0.5 to 2 mcg/kg/minute or general anesthetic

Dependent on duration of procedure

Assisted or controlled ventilation required.

Discontinue at least 30 minutes prior to end of surgery.

Monitored Anesthesia Care (MAC), analgesia

3 to 8 mcg/kg

3 to 5 mcg/kg every 5 to 20 minutes

0.25 to 1 mcg/kg/minute

3 to 40 mcg/kg

Sedation, responsiveness, spontaneously breathing.

May continue to the end of procedure.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; use with caution. The pharmacokinetics of alfentanil were evaluated in adult patients with chronic renal failure and compared to patients with normal renal function. Although V_{dss} was increased in patients with renal failure, elimination half-life was similar between the 2 groups (Ref). Therefore, prolongation of alfentanil duration of action is not expected and dosage adjustment is not necessary.

Dosing: Liver Impairment: Adult

There are no specific dosage adjustments provided in the manufacturer's labeling; use with caution and reduce the dose as needed; monitor closely.

Dosing: Obesity: Adult

In patients weighing >20% above ideal body weight, determine dose based on lean body weight.

Dosing: Older Adult

Refer to adult dosing. Reduce the initial dose by up to 40%; consider the effect of the initial dose in determining supplemental doses.

Dosing: Pediatric

(For additional information see "Alfentanil (United States and Canada: Not available): Pediatric drug information")

Dosage guidance:

Safety: Doses should be titrated to appropriate effects; wide range of doses is dependent upon desired degree of analgesia/anesthesia. Alfentanil should only be used by clinicians trained in the provision of anesthesia to pediatric patients.

Anesthesia

Anesthesia:

Children <12 years:

Pre-induction, emergence agitation prevention, analgesia in tonsillectomy, or dental procedure patients undergoing general anesthesia: Limited data available: IV: 10 to 20 mcg/kg/dose (Ref).

Procedural analgesia for lumbar puncture or bone marrow aspiration (in addition to propofol): Limited data available: Intermittent IV: 2 to 3 mcg/kg/dose (total dose: mean: 1.4 mcg/kg $\pm$ 2.4; range: 1.8 to 9.6 mcg/kg) administered to 20 patients ages 2 to 16 years (Ref).

Children $\geq$ 12 years and Adolescents: IV: See table; **Note:** Base dose on actual body weight unless >20% above ideal body weight, then base dose on lean body weight.

Alfentanil Dosing in Pediatric Patients $\geq$ 12 Years

Indication

Approximate Duration of Anesthesia (minutes)

Induction Period (Initial Dose) (mcg/kg)

Maintenance Period (Increments/ Infusion)

Total Dose (mcg/kg)

Effects

Incremental IV injection

$\leq$ 30 minutes

8 to 20 mcg/kg

3 to 5 mcg/kg every 5 to 20 minutes **or**

0.5 to 1 mcg/kg/minute

8 to 40 mcg/kg

Spontaneously breathing or assisted ventilation when required.

30 to 60 minutes

20 to 50 mcg/kg

5 to 15 mcg/kg every 5 to 20 minutes

Up to 75 mcg/kg

Assisted or controlled ventilation required. Attenuation of response to laryngoscopy and intubation. Discontinue at least 30 minutes prior to end of surgery.

Continuous IV infusion

≥45 minutes

50 to 75 mcg/kg

0.5 to 3 mcg/kg/minute;

average infusion rate: 1 to 1.5 mcg/kg/minute

Dependent on duration of procedure

Assisted or controlled ventilation required. Some attenuation of response to intubation and incision, with intraoperative stability.

Anesthetic induction

≥45 minutes

130 to 245 mcg/kg

0.5 to 1.5 mcg/kg/minute

or general anesthetic

Dependent on duration of procedure

Assisted or controlled ventilation required. Administer induction dose slowly (over 3 minutes). Concentration of volatile inhalation anesthetics reduced by 30% to 50% for initial hour.

Following an anesthetic induction dose of alfentanil, alfentanil IV infusion rate requirements are reduced by 30% to 50% for the first hour of maintenance.

Monitored anesthesia care (MAC)

3 to 8 mcg/kg

3 to 5 mcg/kg every 5 to 20 minutes **or** 0.25 to 1 mcg/kg/minute

3 to 40 mcg/kg

Sedation, responsiveness, spontaneously breathing.

May be continued to the end of the procedure.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. The pharmacokinetics of alfentanil were evaluated in adult patients with chronic renal failure and compared to patients with normal renal function. Although V_{dss} was increased in patients with renal failure, elimination half-life was similar between the 2 groups (Ref). Therefore, prolongation of alfentanil duration of action is not expected and dosage adjustment is not necessary.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use with caution and reduce the dose as needed; monitor closely.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Cardiovascular: Bradycardia (14%), chest wall rigidity (17%), hypertension (18%), tachycardia (12%)

Gastrointestinal: Nausea (28%), vomiting (18%)

1% to 10%:

Cardiovascular: Cardiac arrhythmia (1% to 3%), hypotension (10%)

Dermatologic: Pruritus ($\leq 1\%$), urticaria ($\leq 1\%$)

Local: Pain at injection site ($\leq 1\%$)

Nervous system: Confusion (postoperative; $\leq 1\%$), dizziness (3% to 9%), drowsiness ($\leq 3\%$), euphoria (postoperative; $\leq 1\%$), headache ($\leq 1\%$), sedated state ($\leq 3\%$; postoperative), shivering ($\leq 1\%$)

Neuromuscular & skeletal: Laryngospasm ($\leq 1\%$), muscle movements (3% to 9%; skeletal)

Ophthalmic: Blurred vision (1% to 3%)

Respiratory: Apnea (3% to 9%), hypercapnia ($\leq 1\%$), respiratory depression (1% to 3%; postoperative)

<1%: Respiratory: Bronchospasm

Frequency not defined: Nervous system: Drug abuse, opioid dependence

Postmarketing:

Hypersensitivity: Anaphylaxis (Pepys 1994)

Nervous system: Agitation, allodynia (opioid-induced hyperalgesia) (FDA Safety Communication 2023), epileptiform seizure (Ross 2001), myoclonus

Neuromuscular & skeletal: Muscle rigidity (Benthuisen 1986)

Ophthalmic: Miosis (Black 1999)

Respiratory: Bradypnea, hypoxia

Contraindications

Hypersensitivity (eg, anaphylaxis) to alfentanil or any component of the formulation.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

- Esophageal dysfunction: Opioid-induced esophageal dysfunction has been reported; risk increases with higher doses and prolonged therapy.
- Hyperalgesia: Opioid-induced hyperalgesia (OIH) has occurred with short-term and prolonged use of opioid analgesics. Symptoms may include increased levels of pain upon opioid dosage increase, decreased levels of pain upon opioid dosage decrease, or pain from ordinarily nonpainful stimuli; symptoms may be suggestive of OIH if there is no evidence of underlying disease progression, opioid tolerance, opioid withdrawal, or addictive behavior. Consider decreasing the current opioid dose or opioid rotation in patients who experience OIH.
- Hypotension: May cause hypotension; use with caution in patients with hypovolemia, cardiovascular disease (including acute MI), or drugs which may exaggerate hypotensive effects (including phenothiazines or general anesthetics). Monitor for symptoms of hypotension following initiation or dose titration.
- Hypersensitivity: Anaphylaxis reactions may occur.

- **Respiratory depression:** Fatal respiratory depression may occur. Carbon dioxide retention from opioid-induced respiratory depression can exacerbate the sedating effects of opioids. Have resuscitative and intubation equipment, oxygen, and an opioid overdose reversal agent (eg, naloxone, nalmefene) readily available.
- **Serotonin syndrome:** Potentially life-threatening serotonin syndrome (SS) has occurred with concomitant use of alfentanil and serotonergic agents (eg, SSRIs, SNRIs, triptans, TCAs, 5-HT₃ receptor antagonists, mirtazapine, trazodone, tramadol) and agents that impair metabolism of serotonin (eg, MAO inhibitors). Monitor patients closely for signs of SS such as mental status changes (eg, agitation, hallucinations, delirium, coma); autonomic instability (eg, tachycardia, labile blood pressure, diaphoresis); neuromuscular changes (eg, tremor, rigidity, myoclonus); GI symptoms (eg, nausea, vomiting, diarrhea); and/or seizures. Discontinue alfentanil if serotonin syndrome is suspected.
- **Opioid agonist toxicities:** Shares the toxic potentials of opioid agonists, and precautions of opioid agonist therapy should be observed.

Disease-related concerns:

- **Abdominal conditions:** May obscure diagnosis or clinical course of patients with acute abdominal conditions.
- **Biliary tract impairment:** Use with caution in patients with biliary tract dysfunction, including acute pancreatitis; opioids may cause constriction of sphincter of Oddi.
- **Bradyarrhythmias:** Bradycardia may occur; use with caution when administering to patients with bradyarrhythmias. Degree of bradycardia may be more pronounced when administered with non-vagolytic skeletal muscle relaxants (eg, vecuronium, cisatracurium) or when anticholinergic agents (eg, atropine) are not used.
- **Delirium tremens:** Use with caution in patients with delirium tremens.
- **Head trauma:** Use with extreme caution in patients with head injury, intracranial lesions, or elevated intracranial pressure (ICP); exaggerated elevation of ICP may occur.
- **Hepatic impairment:** Use with caution in patients with hepatic impairment; reduce the dose as needed; monitor closely.
- **Obesity:** Use with caution in patients who are morbidly obese. Reduce dose; use lean body weight for dosing in patients >20% over ideal body weight.
- **Prostatic hyperplasia/urinary stricture:** Use with caution in patients with prostatic hyperplasia and/or urinary stricture.
- **Psychosis:** Use with caution in patients with toxic psychosis.
- **Renal impairment:** Use with caution in patients with renal impairment.
- **Respiratory disease:** Use with caution and monitor for respiratory depression in patients with significant chronic obstructive pulmonary disease or cor pulmonale, and those with a substantially decreased respiratory reserve, hypoxia, hypercapnia, or preexisting respiratory depression, particularly when initiating and titrating therapy; critical respiratory depression may occur, even at therapeutic dosages.
- **Seizures:** Use with caution in patients with a history of seizure disorders; may increase risk or exacerbate preexisting seizure disorders.
- **Skeletal muscle rigidity:** May produce muscular rigidity that involves all skeletal muscles, including those of the neck and extremities; incidence is dose-related. Initial doses up to 20 mcg/kg may cause skeletal muscle rigidity, particularly of the truncal muscles. Doses >130 mcg/kg will consistently cause muscle rigidity with an immediate onset. Consider the concomitant use of a nondepolarizing skeletal muscle relaxant to decrease the incidence.
- **Sleep-related disorders:** Opioid use increases the risk for sleep-related disorders (eg, central sleep apnea, hypoxemia) in a dose-dependent fashion; use with caution.

- Thyroid dysfunction: Use with caution in patients with thyroid dysfunction.

Special populations:

- Cachectic or debilitated patients: Use with caution in cachectic or debilitated patients; there is a greater potential for critical respiratory depression, even at therapeutic dosages. Appropriately reduce the initial dose; consider the effect of the initial dose in determining supplemental doses.
- Older adult: Use with caution in older adults; may be more sensitive to adverse effects. Appropriately reduce the initial dose; consider the effect of the initial dose in determining supplemental doses. Plasma clearance of alfentanil may be reduced and postoperative recovery may be prolonged.

Other warnings/precautions:

- Discontinuation of therapy: Discontinue infusion at least 30 minutes prior to the end of surgery during general anesthesia; during administration for Monitored Anesthesia Care (MAC), infusions may be continued to the end of the procedure.
- Trained individuals: Alfentanil should be administered health care providers specifically trained in the use of anesthetic agents and should not be used in diagnostic or therapeutic procedures outside the monitored anesthesia setting; opioid antagonist, resuscitative and intubation equipment should be readily available.

Product Availability

Alfentanil has been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous [preservative free]:

Generic: 1000 mcg/2 mL (2 mL [DSC]); 2500 mcg/5 mL (5 mL [DSC])

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Alfentanil HCl Intravenous)

1000 mcg/2 mL (per mL): \$5.26

2500 mcg/5 mL (per mL): \$3.77

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Controlled Substance

C-II

Administration: Adult

IV: Administer IV slowly over 3 minutes or by IV continuous infusion.

Preparation for Administration: Adult

May dilute in NS, D5W, D5NS, or LR to a concentration of 25 to 80 mcg/mL.

Usual Infusion Concentrations: Adult

IV infusion: 10 mg in 250 mL (total volume) (concentration: 40 **mcg**/mL) of D₅W or NS

Administration: Pediatric

Parenteral: IV: Inject slowly over 3 to 5 minutes or by continuous IV infusion; in neonates, monitor closely for chest wall rigidity.

Preparation for Administration: Pediatric

Parenteral:

IV: Dilute prior to administration with NS, D5W, D5NS, or LR to a concentration of 25 to 80 mcg/mL.

Continuous IV infusion: Further dilute in D5W, NS, D5NS, or LR to a concentration of 40 mcg/mL.

Storage/Stability

Store intact vials at 20°C to 25°C (68°F to 77°F). Protect from light.

Use: Labeled Indications

Analgesia: Analgesic adjunct for the maintenance of anesthesia with barbiturate/nitrous oxide/oxygen; analgesic with nitrous oxide/oxygen in the maintenance of general anesthesia; analgesic component for monitored anesthesia care.

Anesthetic: Primary anesthetic for induction of anesthesia in general surgery when endotracheal intubation and mechanical ventilation are required.

Medication Safety Issues

Sound-alike/look-alike issues:

Alfentanil may be confused with Anafranil, fentaNYL, remifentanil, SUFentanil

Alfenta may be confused with Sufenta

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (opioids, all formulations and routes of administration) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024; ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021; High-Alert Medications in Long-Term Care (LTC) Settings ©ISMP 2021).

Metabolism/Transport Effects

Substrate of CYP3A4 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: May increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of Opioid Agonists. *Risk X: Avoid*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Alvimopan: Opioid Agonists may increase adverse/toxic effects of Alvimopan. This is most notable for patients receiving long-term (i.e., more than 7 days) opiates prior to alvimopan initiation. Management: Alvimopan is contraindicated in patients receiving therapeutic doses of opioids for more than 7 consecutive days immediately prior to alvimopan initiation. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amphetamines: May increase analgesic effects of Opioid Agonists. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

Buprenorphine: May decrease therapeutic effects of Opioid Agonists. Management: Seek alternatives to buprenorphine in patients receiving pure opioid agonists. If combined in certain pain management situations (eg, surgery), monitor for symptoms of therapeutic failure/high dose requirements or opioid withdrawal symptoms. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ceritinib: ALfentanil may increase bradycardic effects of Ceritinib. Ceritinib may increase serum concentrations of ALfentanil. Management: If use of alfentanil and ceritinib is necessary, consider dosage reduction of alfentanil until stable drug effects are achieved. Frequently monitor patients for respiratory depression, sedation, and bradycardia when these agents are combined. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and chlorpromazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Cimetidine: May increase serum concentrations of ALfentanil. *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of ALfentanil. Management: If concomitant use of alfentanil and moderate CYP3A4 inducers is necessary, consider dosage increase

of alfentanil until stable drug effects are achieved. Monitor patients for signs of opioid withdrawal. *Risk D: Consider Therapy Modification*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Alfentanil. Management: If concomitant use of alfentanil and strong CYP3A4 inducers is necessary, consider dosage increase of alfentanil until stable drug effects are achieved. Monitor patients for signs of opioid withdrawal. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of Alfentanil. Management: If use of alfentanil and moderate CYP3A4 inhibitors is necessary, consider dosage reduction of alfentanil until stable drug effects are achieved. Frequently monitor patients for respiratory depression and sedation when these agents are combined. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Alfentanil. Management: If use of alfentanil and strong CYP3A4 inhibitors is necessary, consider dosage reduction of alfentanil until stable drug effects are achieved. Frequently monitor patients for respiratory depression and sedation when these agents are combined. *Risk D: Consider Therapy Modification*

Desmopressin: Opioid Agonists may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diuretics: Opioid Agonists may increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

Eluxadoline: Opioid Agonists may increase constipating effects of Eluxadoline. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gastrointestinal Agents (Prokinetic): Opioid Agonists may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): May decrease therapeutic effects of Opioid Agonists. Opioid Agonists may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methotrimeprazine: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and methotrimeprazine when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: Opioid Agonists may increase serotonergic effects of Monoamine Oxidase Inhibitors. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nalfurafine: Opioid Agonists may increase adverse/toxic effects of Nalfurafine. Opioid Agonists may decrease therapeutic effects of Nalfurafine. *Risk C: Monitor*

Nalmefene: May decrease therapeutic effects of Opioid Agonists. Management: Avoid the concomitant use of oral nalmefene and opioid agonists. Discontinue oral nalmefene 1 week prior to any anticipated use of opioid agonists. If combined, larger doses of opioid agonists will likely be required. *Risk D: Consider Therapy Modification*

Naltrexone: May increase or decrease therapeutic effects of Opioid Agonists. Management: Seek therapeutic alternatives to opioids. See full drug interaction monograph for detailed recommendations. *Risk X: Avoid*

Nefazodone: Opioid Agonists (metabolized by CYP3A4) may increase serotonergic effects of Nefazodone. This could result in serotonin syndrome. Nefazodone may increase serum concentrations of Opioid Agonists (metabolized by CYP3A4). Management: If concomitant use of opioid agonists that are metabolized by CYP3A4 and nefazodone is necessary, consider dose reduction of the opioid until stable drug effects are achieved. Monitor for increased opioid effects and serotonin syndrome/serotonin toxicity. *Risk D: Consider Therapy Modification*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opioids (Mixed Agonist / Antagonist): May decrease analgesic effects of Opioid Agonists. Management: Seek alternatives to mixed agonist/antagonist opioids in patients receiving pure opioid agonists, and monitor for symptoms of therapeutic failure/high dose requirements (or withdrawal in opioid-dependent patients) if patients receive these combinations. *Risk X: Avoid*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pegvisomant: Opioid Agonists may decrease therapeutic effects of Pegvisomant. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Propofol: Alfentanil may increase adverse/toxic effects of Propofol. Specifically the development of opisthotonus (severe hyperextension and spasticity resulting in arching or bridging position) and/or tonic clonic seizures. *Risk C: Monitor*

Ramosetron: Opioid Agonists may increase constipating effects of Ramosetron. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Samidorphan: May decrease therapeutic effects of Opioid Agonists. Specifically, the risk for opioid withdrawal may be increased. *Risk X: Avoid*

Serotonergic Agents (High Risk): Opioid Agonists (metabolized by CYP3A4) may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Somatostatin Analogs: May increase or decrease analgesic effects of Opioid Agonists. *Risk C: Monitor*

Succinylcholine: May increase bradycardic effects of Opioid Agonists. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Tilidine: May increase therapeutic effects of Opioid Agonists. *Risk X: Avoid*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Zolpidem: May increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and zolpidem when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Long-term opioid use may cause secondary hypogonadism, which may lead to sexual dysfunction or infertility (Brennan 2013).

Pregnancy Considerations

Alfentanil crosses the placenta (Cartwright 1989; Gepts 1986).

Prolonged use of opioids during pregnancy can cause neonatal withdrawal syndrome, which may be life-threatening if not recognized and treated according to protocols developed by neonatology experts. If opioid use is required for a prolonged period in a pregnant woman, advise the patient of the risk of neonatal opioid withdrawal syndrome and ensure that appropriate treatment will be available. Opioids may cause respiratory depression and psychophysiologic effects in the neonate; newborns of mothers receiving opioids during labor should be monitored.

The pharmacokinetic properties of alfentanil are not influenced by pregnancy when administered prior to delivery (Gepts 1986). Alfentanil has been evaluated for use in obstetrical analgesia (Mattingly 2003); other agents are more commonly used (ACOG 209 2019).

The ACOG recommends that pregnant women should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

Alfentanil is present in breast milk (Giesecke 1985).

Breast milk was sampled in nine nonbreastfeeding women undergoing tubal ligation. Alfentanil 50 mcg/kg IV was administered following intubation and additional doses given as needed during the procedure. Alfentanil was present in breast milk 4 hours after the surgery and no longer present in breast milk 28 hours after surgery (Giesecke 1985).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother; monitor infants for excess sedation and respiratory depression. The Academy of Breast Feeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Respiratory and cardiovascular status, blood pressure, heart rate; continue to monitor well after surgery because of the risk for delayed effects; signs or symptoms of opioid-induced esophageal dysfunction (eg, chest pain [noncardiac], dysphagia, regurgitation).

Mechanism of Action

Binds with stereospecific receptors at many sites within the CNS, increases pain threshold, alters pain perception, inhibits ascending pain pathways; is an ultra short-acting opioid

Pharmacokinetics (Adult Data Unless Noted)

Note: An early study of continuous infusion suggested nonlinear pharmacokinetics in neonates (Wiest 1991).

Onset of action: Rapid, within 5 minutes

Duration (dose dependent): 30 to 60 minutes

Distribution: V_d :

Newborns (premature): 0.5 to 0.6 L/kg (Davis 1988; Marlow 1990)

Children: 0.163 to 0.4 L/kg (Davis 1989; Meistelman 1987)

Adults: 0.4 to 1 L/kg

Protein binding:

Neonates: 67%

Adults: 88% to 92%

Bound to α_1 -acid glycoprotein

Metabolism: Hepatic

Half-life elimination:

Newborns (premature): 5.33 to 9 hours (Davis 1988; Marlow 1990)

Children: 40 to 63 minutes (Davis 1988; Meistelman 1987; Roure 1987)

Adults: 90 to 111 minutes

Excretion: Only 1% of dose is excreted unchanged; urine (major route of elimination of metabolites)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Hepatic function impairment: Reduced plasma clearance and extended terminal elimination may develop.

Older adult: Reduced plasma clearance and extended terminal elimination may develop.

Patient Counseling Points

(For additional information see "Alfentanil (United States and Canada: Not available): Patient drug information")

What is this drug used for?

- It is used to put you to sleep for surgery.
- It is used to manage pain during surgery or a procedure.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Feeling dizzy or sleepy
- Upset stomach or throwing up

- Muscle stiffness

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Low blood sugar like dizziness, headache, feeling sleepy, feeling weak, shaking, a fast heartbeat, confusion, hunger, or sweating
- Trouble breathing, slow breathing, or shallow breathing
- Noisy breathing
- Breathing problems during sleep (sleep apnea)
- Fast, slow, or abnormal heartbeat
- Feeling confused
- Chest pain or pressure
- Trouble controlling body movements
- Seizures
- Severe bowel problem like severe constipation or stomach pain
- Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or severe headache
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

Find brand name(s) by country or region (for United States and Canada see separate Brand Names sections)

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Rapifen;
- (AT) Austria: Alfentanil hameln | Rapifen;
- (AU) Australia: Alfentanil hameln;

- (BE) Belgium: Rapifen;
- (BR) Brazil: Alfenta | Rapifen;
- (CO) Colombia: Alfentanil;
- (CZ) Czech Republic: Rapifen;
- (DE) Germany: Alfentanil hameln | Rapifen;
- (EE) Estonia: Rapifen;
- (ES) Spain: Fanaxal | Limifen;
- (FI) Finland: Alfentanil hameln | Alfentanil kalceks | Rapifen | Rapifen orion;
- (FR) France: Alfentanil kalceks | Rapifen;
- (GB) United Kingdom: Alfentanil | Rapifen;
- (GR) Greece: Rapifen;
- (IE) Ireland: Rapifen;
- (IT) Italy: Fentalim;
- (KR) Korea, Republic of: Alfenil | Alfentanil | Hana alfentanil | Spenal;
- (KW) Kuwait: Rapifen;
- (LB) Lebanon: Rapifen;
- (LT) Lithuania: Rapifen;
- (LU) Luxembourg: Rapifen;
- (LV) Latvia: Alfentanil kalceks | Rapifen;
- (MA) Morocco: Rapifen;
- (MY) Malaysia: Rapifen;
- (NL) Netherlands: Alfentanil hameln;
- (NO) Norway: Alfentanil hameln | Alfentanil kalceks | Rapifen;
- (NZ) New Zealand: Alfentanil | Rapifen;
- (PL) Poland: Rapifen;
- (PT) Portugal: Alfentanil | Rapifen;
- (PY) Paraguay: Rapifen;
- (SA) Saudi Arabia: Rapifen;
- (SE) Sweden: Alfentanil hameln | Alfentanil kalceks | Rapifen;
- (SI) Slovenia: Alfentanil hameln;
- (SK) Slovakia: Rapifen;
- (TN) Tunisia: Rapifen;
- (TR) Turkey: Rapifen;
- (TW) Taiwan: Alfentanil hameln | Rapifen;
- (VE) Venezuela, Bolivarian Republic of: Rapifen;
- (ZA) South Africa: Rapifen

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